

# Electrical Bench Layout + Test Guide

As-of date: 2026-07-19

Purpose: provide a practical, print-friendly game plan for building, hard-mounting, post-live cleanup, and staged validation of the electrical module before mobile install or permanent enclosure/panel closeout.

Related docs:

- docs/core/ELECTRICAL\_48V\_ARCHITECTURE.md
- docs/implementation/ELECTRICAL\_overview\_diagram.md
- docs/implementation/ELECTRICAL\_fuse\_schedule.md
- docs/plans/PROJECT\_build\_order\_of\_operations.md

## 1) Build intent lock (this guide assumes)

- Architecture remains 48V core + 12V distribution + shore/inverter AC .
- Alternator path baseline is the dedicated 48V secondary alternator path ( Mechman + WS500 + APM-48 ).
- Electrical closet is a two-plane vertical corner build:
  - Board A:** against bed wall (high-current 48V hardware).
  - Board B:** against bulkhead (controls, 12V , AC support hardware).
- Closet sits above battery zone.
- Structural mounting intent is through-bolts with washers and lock nuts (no wood screws for permanent component mounting).
- The installed camper shell owns final outer dimensions and service-access geometry; garage/floor layouts are templates only until the two-plane mockup passes in-camper insertion, removal, cable-bend, fuse-access, and panel-access checks.

## 2) Recommended sequence (ground build -> test fit -> hard mount)

Current owner status ( 2026-07-19 ): the electrical module has passed the first-live checkpoint and is now hard-mounted through the finished floor to truck-bed hardpoints. The MultiPlus and combined AC breaker enclosure were removed for the lift and will remount into embedded pronged T-nuts. All three battery branch harnesses are cut, lugged, heat-shrunk, and landed at the battery-side busbars, but the batteries remain isolated pending individual charge/rest/voltage matching.

Immediate priority:

- Remount the MultiPlus and AC breaker enclosure; prove/protect the full L5-30/ 10/3 shore-inlet path before cutting and energizing it.
- Charge Batteries 2 and 3 individually, rest/record all three voltages, and parallel only when maximum-to-minimum difference is  $\leq 0.1V$  .
- Locate/install final Orion F-06 20A/80V ; identify any X-marked Lynx fuse by slot ( F-03 60A/80V in Slot 2, Slot 4 empty), then complete the 12V battery branch and prove factory loads.
- Preserve the current working topology; do not add alternator or AC-out branch complexity until these cleanup checks pass.
- Add/verify labels, covers, torque witness marks, J-clamp strain relief, cable protection, and service access around the energized layout.
- Keep AC-out branch/GFCI validation and Mechman/WS500 alternator commissioning as separate later gates.

### Phase L1: Build the two backer boards on the ground

- Rip the 1/2" sheet into Board A + Board B to your target envelope.
- Mark orientation on both boards: TOP , BOTTOM , FRONT , REAR , BED WALL , BULKHEAD .
- Place boards at 90 degrees on temporary braces/clamps to represent closet geometry.

Exit criteria:

- Two boards stand in correct orientation and are easy to reposition.

### Phase L2: Paper layout before drilling

- Make 1:1 paper templates for each major component.
- On each template, mark:
  - footprint
  - hole centers
  - cable entry/exit sides
  - keep-out for bend radius + tool access
- Tape templates to boards and pencil trace candidate positions.

4. Reserve explicit channels:
  - high-current 48V trunk path
  - low-current control/comms path
  - AC route path separated from DC path

Exit criteria:

- No blocked fuse access, no blocked terminals, and no impossible cable bends.

### Phase L3: Truck-bed dry fit (before full drilling)

1. Move bare boards (or cardboard stand-ins with same outer dimensions) into truck bed.
2. Validate insertion, rotation, and removal path.
3. Validate service access around door swing/panels.
4. Adjust board size or corner angle now if needed.

Exit criteria:

- Boards can be inserted and removed without forcing or trimming installed components.

### Phase L4: Pilot drill + subset mount

1. Pilot drill only ( 1/8" class) for critical components first.
2. Mount a subset first (example: disconnect, Lynx, shunt, Orion, one AC enclosure).
3. Re-check wrench access, cable swing, and service pull paths.
4. Only after pass, drill final hole sizes and mount remainder.

Exit criteria:

- Critical devices fit and are serviceable with real tools/hands.

### Phase L5: Final mounting + board prep

1. Through-bolt permanent component mounts.
2. Add backing washers/plates where loads are concentrated.
3. Seal plywood faces/edges after hole pattern is validated.
4. Label each component position and cable corridor directly on board.

Exit criteria:

- Boards are mechanically complete and labeled for wiring.

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## 3) Suggested zoning by board

### Board A (bed-wall plane, high-current focus)

- Place heavier/high-current gear lower when possible for stability and short battery paths.
- Keep these grouped for short 48V runs:
  - main disconnect
  - Lynx
  - shunt
  - major 48V feeds/returns
- Keep alternator branch landing ( F-04 path) accessible and clearly labeled.
- Keep Class-T and major fuse service points reachable without removing adjacent equipment.

### Board B (bulkhead plane, controls + distribution)

- Place 12V panel, Orion, monitoring/comms, and AC panel hardware here.
- Keep AC hardware grouped and physically separated from DC/control paths.
- Keep control and comms routing high/clean, away from high-current cable bundles.

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## 4) Wiring layout rules (before you crimp anything)

- Keep parallel battery paths balanced by similar total loop resistance. Equal positive-only leads are not required if short/medium/long positives are offset by long/medium/short negatives.

- Before paralleling, charge/rest/measure each battery separately and require maximum-to-minimum terminal-voltage difference  $\leq 0.1V$  ; do not use the final parallel bus to equalize a larger mismatch.
- Keep 48V high-current paths short and direct.
- Keep AC and DC routing physically separated; cross at 90 degrees only when unavoidable.
- Use abrasion protection at every edge pass-through.
- Leave service loops at removable devices (do not strain to exact length).
- Do not cut final shell-dependent runs yet; only bench-module harnesses at this stage.

## 5) Bench test ladder (staged energization)

Use a stop/go ladder; do not skip steps.

### T0: Dead checks

- Verify polarity labels and continuity on every cable.
- Verify torque marks applied to all completed terminations.
- Verify fuse IDs/ratings match docs/implementation/ELECTRICAL\_fuse\_schedule.md .
- Before any AC energization, meter the AC system dead: AC-in and AC-out neutrals are isolated, PE/equipment ground is continuous through inlet/enclosure/MultiPlus/receptacle paths, and no fixed downstream neutral-ground bond exists.

Pass when:

- No polarity/continuity mistakes and all terminations documented.

### T1: Low-risk controls and comms

- Validate control circuits and monitoring links first ( WS500 control path, comms, panel labeling).
- Confirm Upfitter #3 control path is treated as low-current signal only.

Pass when:

- Control/monitoring paths behave as expected with no unexplained faults.

### T2: 48V core energization (no alternator)

- Bring up house 48V path first.
- Use pre-charge before closing the main disconnect into MultiPlus/inverter capacitance.
- Verify expected voltage at key nodes and no abnormal heat/smell/noise.
- Confirm disconnect behavior and basic inverter/charger DC-side response.

Current result ( 2026-05-27 ):

- Owner measured 55.5V throughout the system, including at the MultiPlus.
- MultiPlus switch I brought inverter mode online; inverter light illuminated, slight hum was observed, and no error lights were reported.
- SmartShunt and Orion-Tr Smart were visible in VictronConnect.

Pass when:

- Stable voltage and normal behavior at all checkpoints, with labels/covers/torque marks verified after the first live session.

### T2.5: Shore / MultiPlus initial charge validation

- Build the AC-in-only path first: shore source/adapters -> portable EMS -> shore cord -> inlet -> AC-in breaker/disconnect -> MultiPlus AC-in.
- Keep AC-out branch breakers/loads disconnected for the first charge test.
- Confirm the T0 AC dead-checks are complete before shore energization.
- Set AC Input 1 label/type to Shore power and set the MultiPlus input current limit to the actual source. Use 10A for first household tests and 12A maximum policy on a normal 15A outlet.
- Leave DVCC disabled unless a documented BMS/GX control path is added.
- Before sustained charging, program/verify MultiPlus LiFePO4 charge profile from the Dumfume manual ( MK3-USB + VEConfigure or equivalent): absorption/float, charge current, equalization off, and lithium temperature-compensation behavior.
- Use the pre-charge procedure before closing the 48V disconnect into MultiPlus capacitance.
- Leave alternator branch inactive and F-04 out during shore-charge validation.
- Log starting/resting voltage, charge current, temperature, SOC/monitor readings, abnormal heat/noise/smell, input-current limit, and stop conditions.

Current result ( 2026-06-01 ):

- Short AC-in test passed at limited household-outlet draw: about 1294W shore input and about 54.3V x 21.6A ( ~1173W ) battery charging in bulk.
- Charge-profile verification is now treated as passed for the current shore-charger setup: owner redid settings and confirmed first-battery behavior entered bulk, then quickly transitioned to absorption because the battery was already at/near 100% , as planned. No second-battery charge is required just to validate the profile.

Pass when:

- Shore charging repeats normally with programmed battery settings, no unexplained BMS trips, abnormal heat, polarity issues, or AC faults, and logged values match expected limits.

### T3: 12V distibution validation

- Validate Orion-fed 12V junction behavior and branch fusing.
- Test representative 12V loads one at a time, then combined.

Pass when:

- 12V rail remains stable under expected test loads.

### T4: AC path validation

- Validate AC-in and AC-out paths in staged order with small known loads first.
- Reconfirm neutral isolation, PE continuity, no downstream neutral-ground bond, and correct outlet polarity before applying known loads.
- Confirm branch protection behavior and test each GFCI receptacle in its installed branch context.

Pass when:

- AC paths behave correctly in shore and inverter modes, with GFCI behavior verified and no neutral/ground faults found.

### T5: Alternator path readiness (pre-engine integration)

- Confirm alternator branch hardware is present/labeled/serviceable.
- Confirm WS500 enable/disable control path labeling and fuse placement ( F-15 ).
- Defer live engine alternator validation until full install context and safety controls are ready.

Pass when:

- Alternator branch is install-ready with shutdown path understood and labeled.

Stop conditions (halt and fix before continuing):

- Any reversed polarity, unexpected voltage, heating at terminals, damaged insulation, nuisance fuse trips, or unclear shutdown behavior.

## 6) Garage workday card (print/use)

Use one card per session:

- **Date:**
- **Board focus:** A / B / Both
- **Phase:** L1 / L2 / L3 / L4 / L5 / T0-T5
- **What changed:**
- **What passed:**
- **What failed or needs rework:**
- **Next single action:**

## 7) Hold points (do not cross yet)

- Do not finalize shell-dependent cable cuts.
- Do not leave shore charging unattended while wiring is loose, unprotected, being rerouted, or missing required strain relief/covers; the MultiPlus profile itself is programmed/owner-verified.
- Do not treat the short AC-in charge test as AC-out branch commissioning; GFCI/outlet validation remains separate.
- Do not lock roof/solar final terminations.
- Do not permanently close access panels before all staged tests pass.
- Do not use main 48V disconnect as first alternator shutdown method while charging is active; use Upfitter #3 ( WS500 disable path ) first.